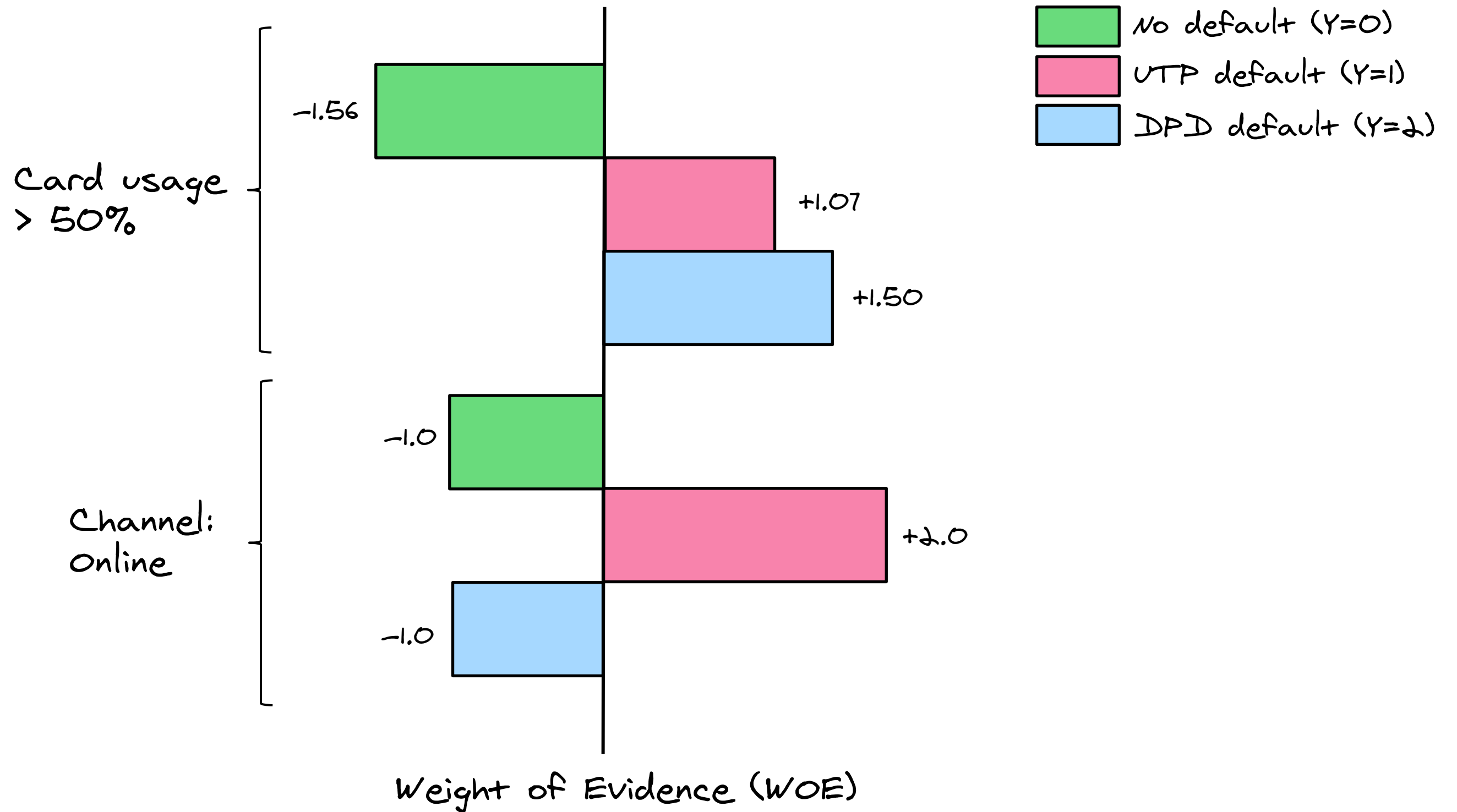


Multiclass Weight of Evidence



Multiclass Weight of Evidence (WOE)

Binary Problem with Simple Hypotheses



Weight of Evidence (WOE) is a statistical technique widely used in credit scoring and other banking data science use cases.

In the binary case, we compare two hypotheses (e.g. Default vs No Default) by calculating a likelihood ratio or log likelihood ratio (WOE).

Using likelihood ratios and WOE, we can say how much evidence from our data increases or decreases the belief in the hypothesis (e.g., default), and when combined with prior odds (e.g., sample default rate) we obtain the probability of default (PD).

In the example, we see that having 5+ days of delinquency increases the odds of default almost 5 times (WOE = +1.6).

The problem is straightforward when we consider only two classes but quickly becomes non-trivial when we have more than two classes, yet the governing principle remains the same, as we show next.

Evidence (X)	Hypotheses		Row Sum
	Default (Y=1)	No Default (Y=0)	
Delinquent > 5 days	80	100	180
Delinquent ≤ 5 days	20	500	520
Column Sum	100	600	700

Evidence (X)	Hypotheses		Likelihood ratio $P(X=j Y=1) \div P(X=j Y=0)^1$
	Default (Y=1)	No Default (Y=0)	
Delinquent > 5 days	0.80	0.167	4.80
Delinquent ≤ 5 days	0.20	0.833	0.24
Column Sum	1.00	1.00	-

$$LR_j = \frac{P(X = j | y = 1)}{P(X = j | y = 0)}$$

Likelihood Ratio

$$WOE_j = \log \left(\frac{P(X = j | y = 1)}{P(X = j | y = 0)} \right)$$

Weight of Evidence

1. $P(X = j | Y)$: likelihood of observing this evidence in group j given each outcome.



Multiclass with Simple Hypotheses

Multiclass Weight of Evidence (WOE)

Multiclass Problem Formulation with Simple Hypotheses



Now imagine that we wanted to look at different types of default, Days Past Due (DPD) and Unlikely to Pay (UTP) through a similar lens¹.

In a multiclass setting, we can't directly compare all three at once with a single ratio. Instead, we must choose a **reference hypothesis** and compute pairwise likelihood ratios against it.

Evidence (X)	Hypotheses			Row Sum
	DPD Default (Y=2)	UTP Default (Y=1)	No Default (Y=0)	
Delinquent > 5 days	50	30	100	180
Delinquent ≤ 5 days	5	15	500	520
Column Sum	55	45	600	700

Evidence (X)	Conditional probabilities, $P(X Y)$			How to get a likelihood ratio?
	DPD Default (Y=2)	UTP Default (Y=1)	No Default (Y=0)	
Delinquent > 5 days	0.91	0.67	0.17	-
Delinquent ≤ 5 days	0.09	0.33	0.83	-
Column Sum	1.00	1.00	1.00	-

Multiclass LR with Simple Hypotheses

$$\begin{array}{ccc}
 \text{LR}_{2|0} & \text{LR}_{1|0} & \text{LR}_{0|0} \\
 \frac{P(X=j | Y=2)}{P(X=j | Y=0)} & \frac{P(X=j | Y=1)}{P(X=j | Y=0)} & \frac{P(X=j | Y=0)}{P(X=j | Y=0)}
 \end{array}$$

For instance, if:

- $P(X > 5 | \text{DPD}) = 0.91$
- $P(X > 5 | \text{No Default}) = 0.17$

then $LR = 0.91/0.17 = 5.4$, $WOE = +1.69$.

This means that odds of DPD default are 5.4 times higher than no default with a more than 5 days delinquency.

$$LR_{i,j|r} = \frac{P(X = j | y = i)}{P(X = j | y = r)}$$

Likelihood Ratio

$$WOE_{i,j|r} = \log \left(\frac{P(X = j | y = i)}{P(X = j | y = r)} \right)$$

Weight of Evidence

1. This example will become part of a chapter dedicated to Logistic Regression with applications in Credit Risk.

Multiclass Weight of Evidence (WOE)

Multiclass Inference with Simple Hypotheses



Evidence (X)	Hypotheses			Row Sum
	DPD Default (Y=2)	UTP Default (Y=1)	No Default (Y=0)	
Delinquent > 5 days	50	30	100	180
Delinquent ≤ 5 days	5	15	500	520
Row sum	55	45	600	700
Prior Odds _{reference}	0.09	0.08	1.00	-

Evidence (X)	Conditional probabilities, P(X Y)			LR _{2 0}
	DPD Default (Y=2)	UTP Default (Y=1)	No Default (Y=0)	
Delinquent > 5 days	0.91	0.67	0.17	5.40
Delinquent ≤ 5 days	0.09	0.33	0.83	0.11
Column Sum	1.00	1.00	1.00	-

Posterior Odds: $O_i(E) = O_i \times LR_{i|0}$

Posterior Log Odds: $\log O_i(E) = \log(O_i) + WOE_{i|0}(E)$

Using the **simple hypotheses**, we can derive posterior odds by multiplying prior odds by a corresponding pairwise likelihood ratio (LR):

DPD Default

- Prior Odds_{2|0} = 55 / 600 ≈ 0.09
- LR_{2|0} = 0.91 / 0.17 ≈ 5.5
- Posterior Odds_{2|0} = 0.09 × 5.5 = 0.50

UTP Default

- Prior Odds_{1|0} = 45 / 600 = 0.08
- LR_{1|0} = 0.67 / 0.17 ≈ 4.0
- Posterior Odds_{1|0} = 0.08 × 4.0 ≈ 0.30

No Default

- Prior Odds_{0|0} = 600 / 600 = 1.0
- LR_{0|0} = 0.17 / 0.17 = 1.0
- Posterior Odds_{0|0} = 1.0 × 1.0 = 1.0

To calculate the PD for DPD events given delinquency of more than 5 days, we can use the Bayes' theorem:

$$PD_{DPD} = 0.50 / (0.50 + 0.30 + 1.0) = 28\%$$



Multiclass with Simple-vs-Composite Hypotheses

Multiclass Weight of Evidence (WOE)

Multiclass Problem Formulation with Simple-vs-Composite Hypotheses



Continuing our example with the types of default, Days Past Due (DPD) and Unlikely to Pay (UTP), we can use a different formulation to derive likelihood ratios and weights of evidence (WOE).

Here, we are only interested in only one hypothesis, ignore the alternative (composite, e.g., not DPD). The key idea is that now we work with row sums to calculate LR's. This is based on the Odds form of the Bayes' Theorem as formulated by I.J. Good¹.

Evidence (X)	Hypotheses			Row Sum
	DPD Default (Y=2)	UTP Default (Y=1)	No Default (Y=0)	
Delinquent > 5 days	50	30	50	180
Delinquent ≤ 5 days	5	15	200	520
Column Sum	55	45	600	700

Evidence (X)	Conditional probabilities, $P(Y X)$			Row Sum
	DPD Default (Y=2)	UTP Default (Y=1)	No Default (Y=0)	
Delinquent > 5 days	0.28	0.17	0.56	1.00
Delinquent ≤ 5 days	0.01	0.03	0.96	1.00
Column Sum	-	-	-	-

$$\underbrace{\frac{O(H|E)}{O(H)}}_{\text{Posterior odds}} = \underbrace{\frac{P(E|H)}{P(E|\bar{H})}}_{\text{Likelihood ratio}}$$

As I.J. Good notes, the same odds-form Bayes formula applies whether the hypotheses are simple (a single, fully specified case) or composite (composed of multiple simple hypotheses). When we focus on one hypothesis H , the general formula derived from the Bayes' Theorem gives us the essential quantity for inference, posterior odds $O(H|E)$.

Multiclass LR with Composite Hypotheses

$$\begin{array}{ccc} \text{LR}_{2|¬2} & \text{LR}_{1|¬1} & \text{LR}_{0|¬0} \\ \left(\frac{P(Y=2 | X=j)}{P(Y \neq 2 | X=j)} \right) & \left(\frac{P(Y=1 | X=j)}{P(Y \neq 1 | X=j)} \right) & \left(\frac{P(Y=0 | X=j)}{P(Y \neq 0 | X=j)} \right) \\ \left(\frac{P(Y=2)}{P(Y \neq 2)} \right) & \left(\frac{P(Y=1)}{P(Y \neq 1)} \right) & \left(\frac{P(Y=0)}{P(Y \neq 0)} \right) \end{array}$$

1. I.J. Good. *Good Thinking: The Foundations of Probability and Its Applications*. University of Minnesota Press, 1983.

Multiclass Weight of Evidence (WOE)

Multiclass Inference with Simple-vs-Composite Hypotheses



Evidence (X)	Hypotheses			Row Sum
	DPD Default (Y=2)	UTP Default (Y=1)	No Default (Y=0)	
Delinquent > 5 days	50	30	100	180
Delinquent ≤ 5 days	5	15	500	520
Row sum	55	45	600	700
P(Y) ¹	0.08	0.06	0.86	1.00
1 – P(Y)	0.92	0.94	0.14	-
Prior Odds _{composite}	0.09	0.07	6.0	-

Evidence (X)	Conditional probabilities, P(Y X)			LR
	DPD Default (Y=2)	UTP Default (Y=1)	No Default (Y=0)	
Delinquent > 5 days	0.28	0.17	0.56	1.00
Delinquent ≤ 5 days	0.01	0.03	0.96	1.00
Column Sum	-	-	-	-

To calculate likelihood ratios, we need to divide odds of Y | X by the prior Y, which gives us an odds ratio

DPD Default

- Prior Odds_{2|¬2} = 0.08 / 0.92 ≈ 0.09
- LR_{2|¬2} = (0.28 / (1 – 0.28)) / 0.09 ≈ 4.51
- Posterior Odds_{2|¬2} = 0.09 × 4.51 ≈ 0.39

UTP Default

- Prior Odds_{1|¬1} = 0.06 / 0.94 ≈ 0.07
- LR_{1|¬1} = (0.17 / (1 – 0.17)) / 0.07 = 2.91
- Posterior Odds_{1|¬1} = 0.07 × 2.91 = 0.20

No Default

- Prior Odds_{0|¬0} = 0.86 / 0.14 ≈ 6.0
- LR_{0|¬0} = (0.56 / (1 – 0.56)) / 6.0 ≈ 0.21
- Posterior Odds_{0|¬0} = 6.14 × 0.21 = 1.25

To calculate the PD for DPD events given delinquency of more than 5 days, we can use odds to probability formula:

$$PD_{DPD} = 0.39 / (1 + 0.39) = 28\%$$

1. For example, DPD prior probability is 55 / 700 = 0.08.



Some open questions:

- What is the connection between multiclass likelihood ratios / WOE and multinomial logistic regression?
- How can we derive posterior class predictions using multiclass LR / WOE?
- Can target encoding be viewed as a practical way to compute multiclass likelihood ratios?

I'll explore these questions (and more) in the next chapter:



Logistic Regression: From Maximum Likelihood to Machine Learning